

Operating System

What is an Operating System?

An **Operating System (OS)** is system software that manages computer hardware, software resources, and provides common services for computer programs. It acts as an **intermediary** between the user and the hardware.

Main functions of an OS:

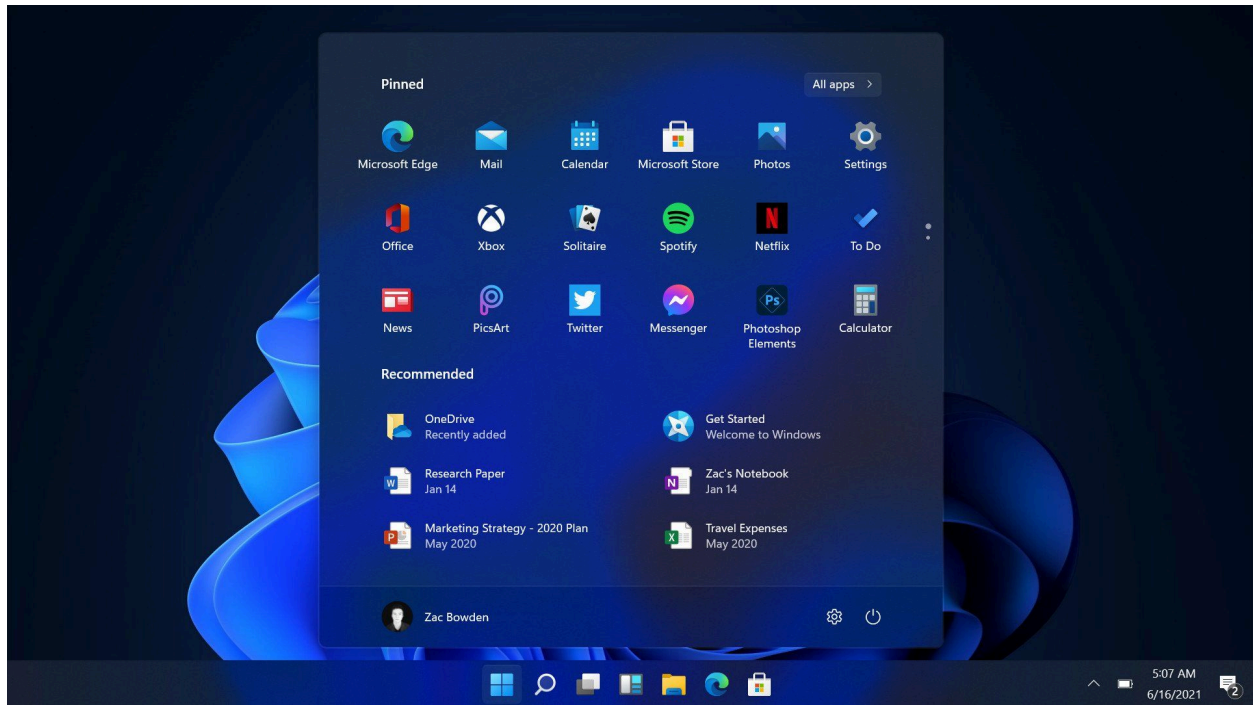
- Process management
- Memory management
- File management
- Device management (keyboard, mouse, printer, etc.)
- User interface
- Security & multitasking

Basics of Popular Operating Systems

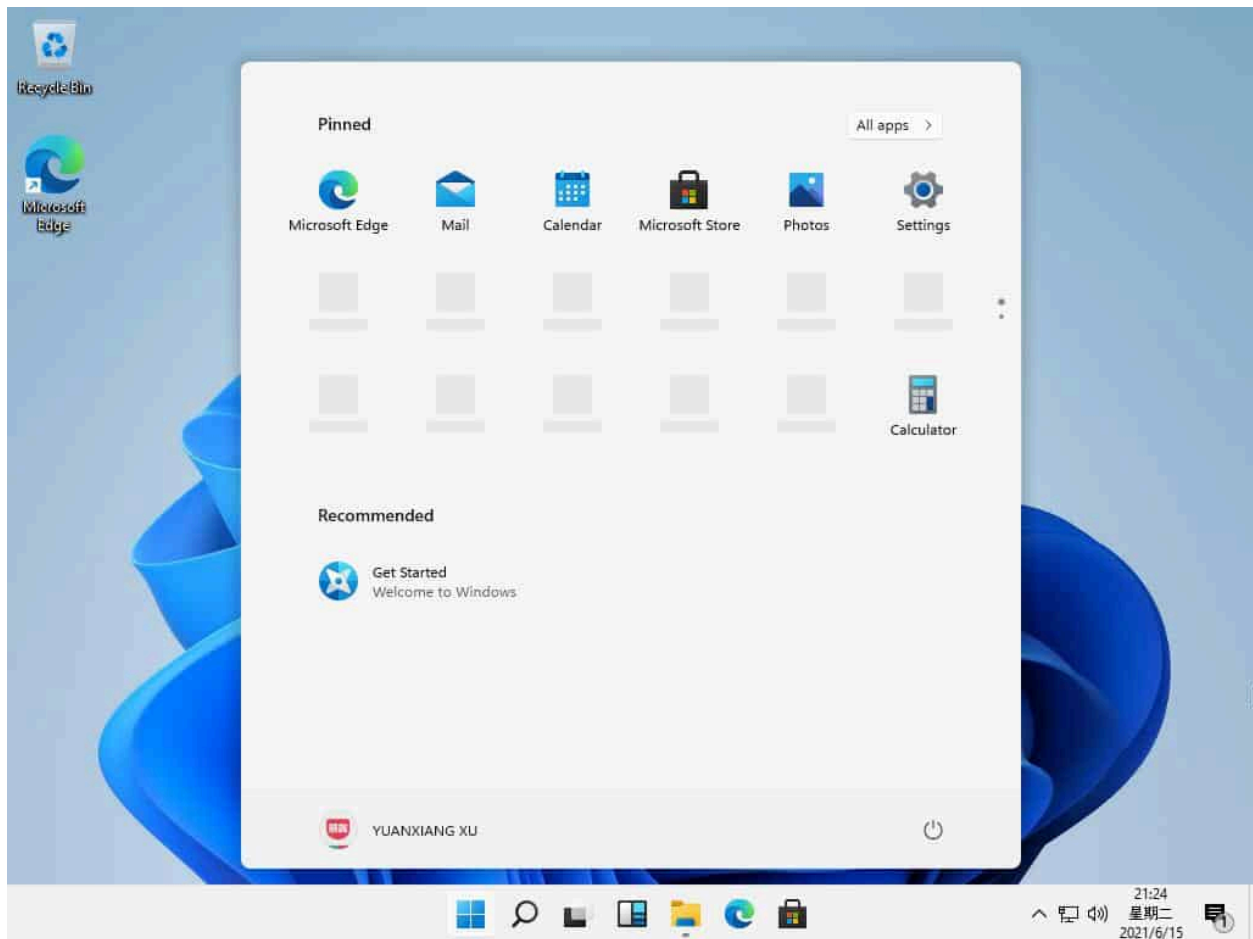
Windows (Microsoft Windows)

- Most widely used desktop OS (user-friendly, GUI-focused)
- Versions: Windows 10 → Windows 11 (current major version in 2026)
- Strengths: Huge software/game compatibility, excellent hardware support, intuitive for beginners
- Here are some examples of the modern **Windows 11** desktop interface (centered taskbar & Start menu):

Here are some visual examples of Windows 11 desktop with Taskbar, Start Menu, and icons:



[windowscentral.com](https://www.windowscentral.com)

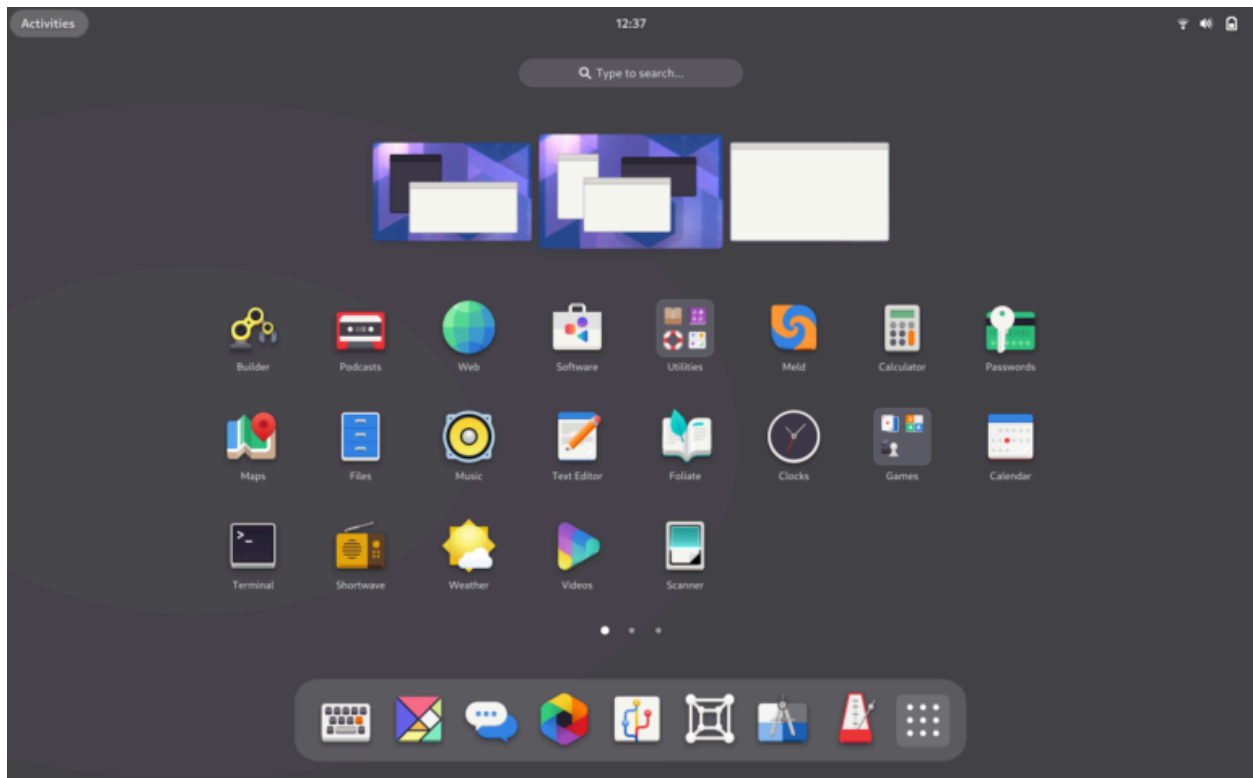


ghacks.net

Linux

- Free & open-source OS (many distributions or "distros")
- Popular distros: Ubuntu, Linux Mint, Fedora, Pop!_OS
- (very beginner-friendly)
- Strengths: Extremely customizable, secure, lightweight, free,
- great for servers/programming
- Most modern Linux desktops use beautiful graphical

Here are examples of a popular **GNOME** Linux desktop (with dock/taskbar equivalent):



discourse.gnome.org

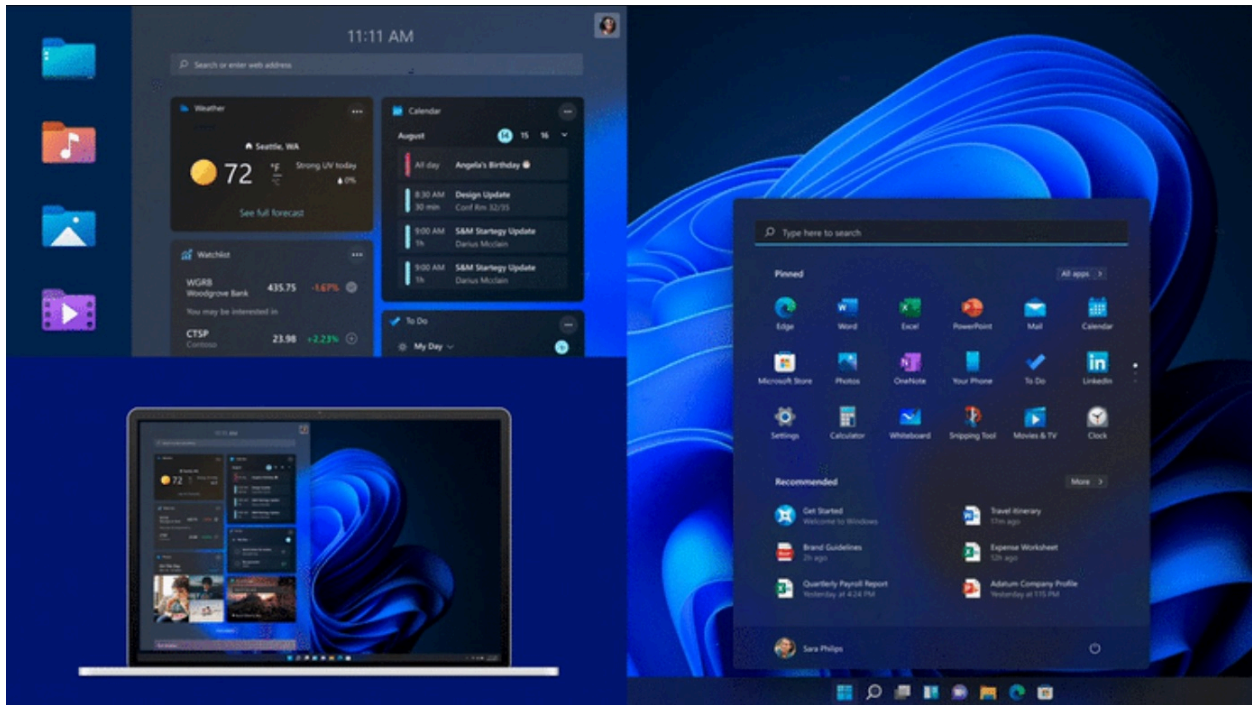
Dock Improvement - Desktop - GNOME Discourse

The User Interface (Windows Focus)

Main Elements in Windows 11

- **Taskbar** — Bottom bar (centered by default); shows open apps,
- pinned apps, system tray (clock, network, volume, etc.)
- **Icons** — Shortcuts on Desktop or pinned to Taskbar/Start
- **Start Menu** — Click Windows logo → shows pinned apps,
- recommended files, search bar, power options
- **Running an Application** — Double-click icon/shortcut,
- search in
- Start, or use Run (Win + R)

Here are some visual examples of Windows 11 Start Menu & Taskbar:



[windowslatest.com](https://www.windowslatest.com)

Microsoft explains Windows 11's new Start Menu and taskbar design

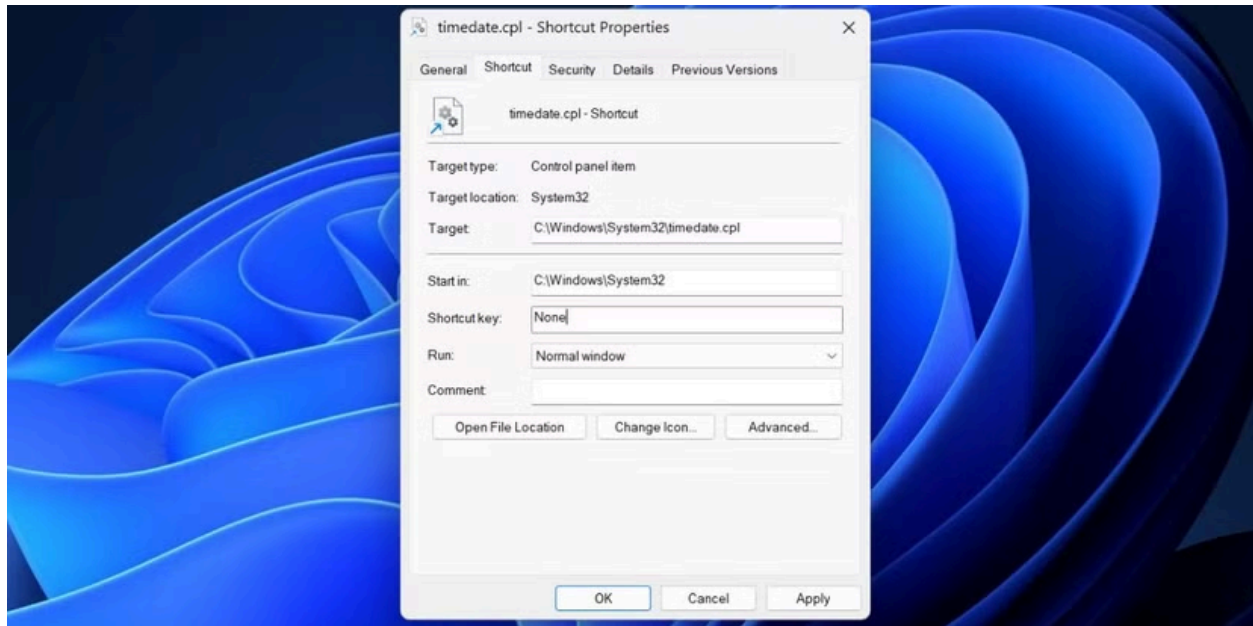
Operating System Simple Settings (Windows)

All settings are accessed mainly via **Settings app** (Win + I key) or right-click → Settings.

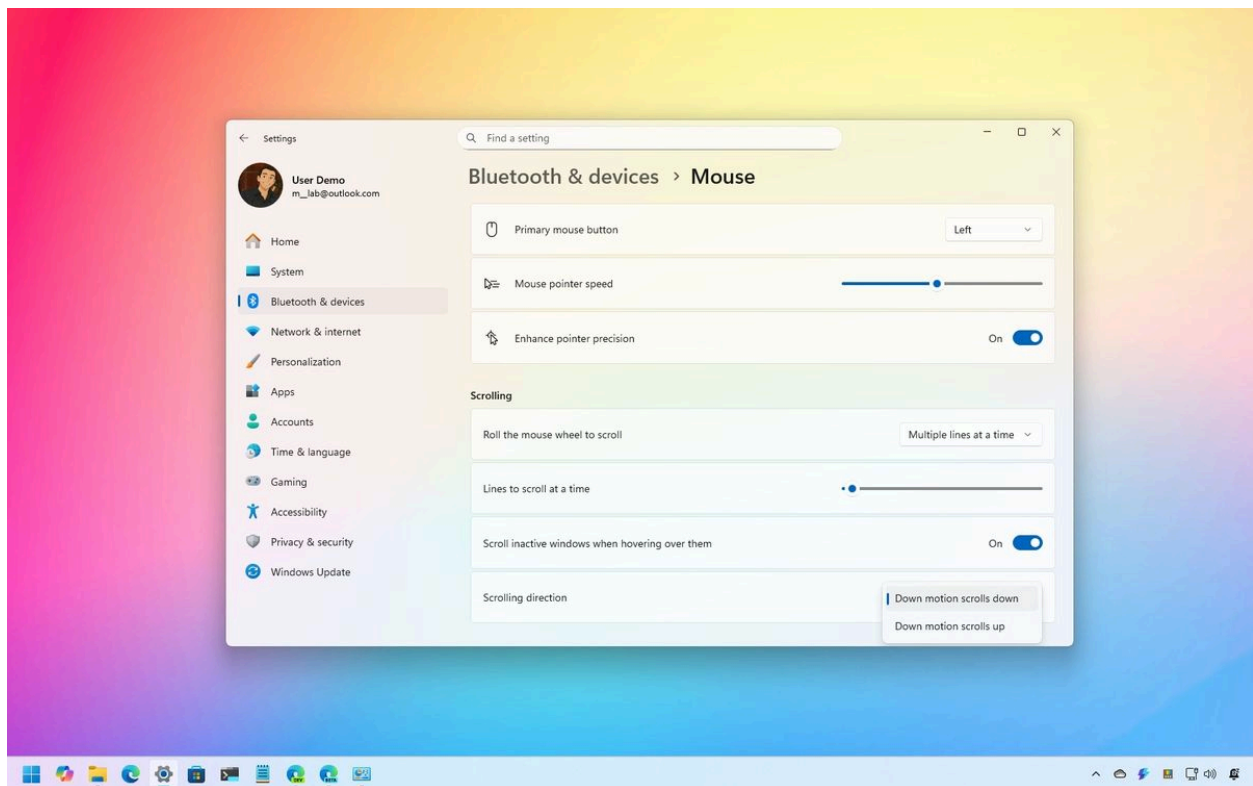
Common simple tasks:

- **Changing System Date and Time** → Settings → Time & language → Date & time
- **Changing Display Properties** → Settings → System → Display (resolution, scale, night light, etc.)
- **Changing Mouse Properties** → Settings → Bluetooth & devices → Mouse (speed, buttons, etc.)

Here are some examples of these settings windows:



makeuseof.com



windowscentral.com

- **Add/Remove Windows Components** → Settings → Apps → Optional features

- **Adding/Removing Printers** → Settings → Bluetooth & devices → Printers & scanners

File and Directory Management (Windows Focus)

What is a File?

A file is a collection of data stored with a name (text document, photo, program, etc.).

Types of Files

- **Document** (.docx, .pdf, .txt)
- **Image** (.jpg, .png, .gif)
- **Video** (.mp4, .mkv)
- **Audio** (.mp3, .wav)
- **Executable/Program** (.exe, .msi)
- **Compressed** (.zip, .rar)

Naming Conventions & File Extensions

- Name can have letters, numbers, spaces, some symbols
- Extension (after the dot) tells Windows what type of file it is (e.g., report.docx)
- Do **not** change extensions manually — it can make files unusable!

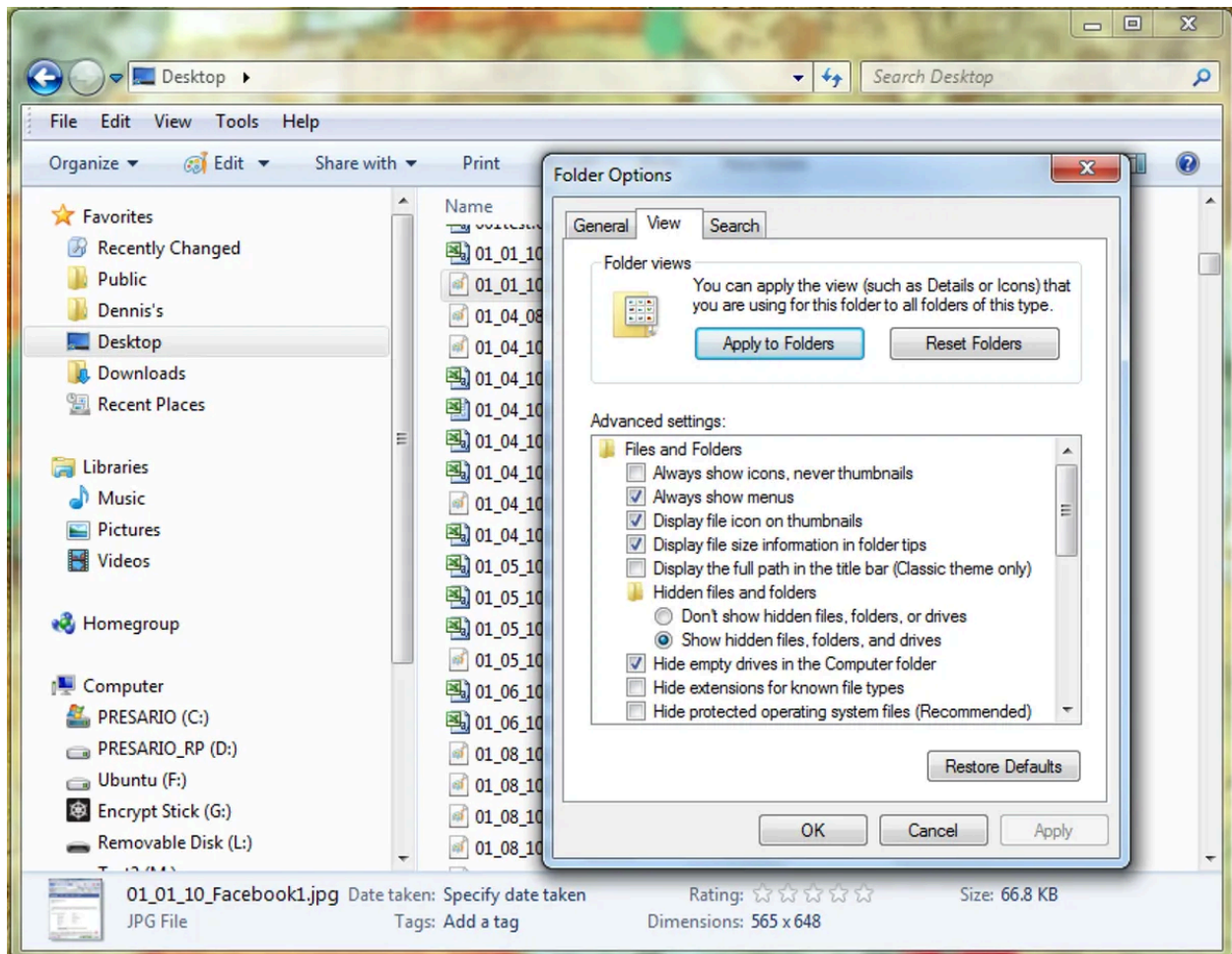
File Pathway / Path

Example: C:\Users\Sam\Documents\Project\report.docx
(Drive → Folders → File)

Windows Explorer (File Explorer)

Main tool to manage files/folders. Open with Win + E.

Here are examples of File Explorer windows showing files, folders, views, and properties:



cnet.com

How to see the same view in all Windows 7 folders - CNET

File Size Units

- **1 byte** = 8 bits (smallest unit)
- **1 KB** (Kilobyte) $\approx$ 1,000 bytes
- **1 MB** (Megabyte) $\approx$ 1,000 KB (1 million bytes)
- **1 GB** (Gigabyte) $\approx$ 1,000 MB (1 billion bytes)
- **1 TB** (Terabyte) $\approx$ 1,000 GB (1 trillion bytes)

Basic File Operations

- **Create a Folder** — Right-click → New → Folder

- **Move files** (multiple) → Select (Ctrl + click or Shift for range) → Cut (Ctrl+X) → Paste (Ctrl+V) in target folder
- **Delete** → Select → Delete key (goes to Recycle Bin)
- **Recover deleted files** → Open Recycle Bin → Right-click file → Restore
- **Rename** → Right-click → Rename (or press F2)
- **Search for files** → Type in File Explorer search box (top right)
- **Create shortcut on desktop** → Right-click file → Send to → Desktop (create shortcut)
- **How programs save files by default** — Many save in **Documents**, **Downloads**, **Pictures**, or **Desktop**. Check "Save As" dialog to see/change location.

Desktop: Exploring & Cleaning

The **Desktop** is your main working area.

Here are examples showing a cluttered desktop vs. a cleaned/organized one:



firstclasstechnologies.co.uk

How to organise a messy Windows desktop - FIRST CLASS TECHNOLOGIES

Tips to clean:

- Move files to proper folders (Documents, Pictures, etc.)
- Create folders on desktop for organization
- Right-click desktop → View → Auto arrange icons / Align icons to grid
- Use Desktop cleanup tools or manually delete unused shortcuts

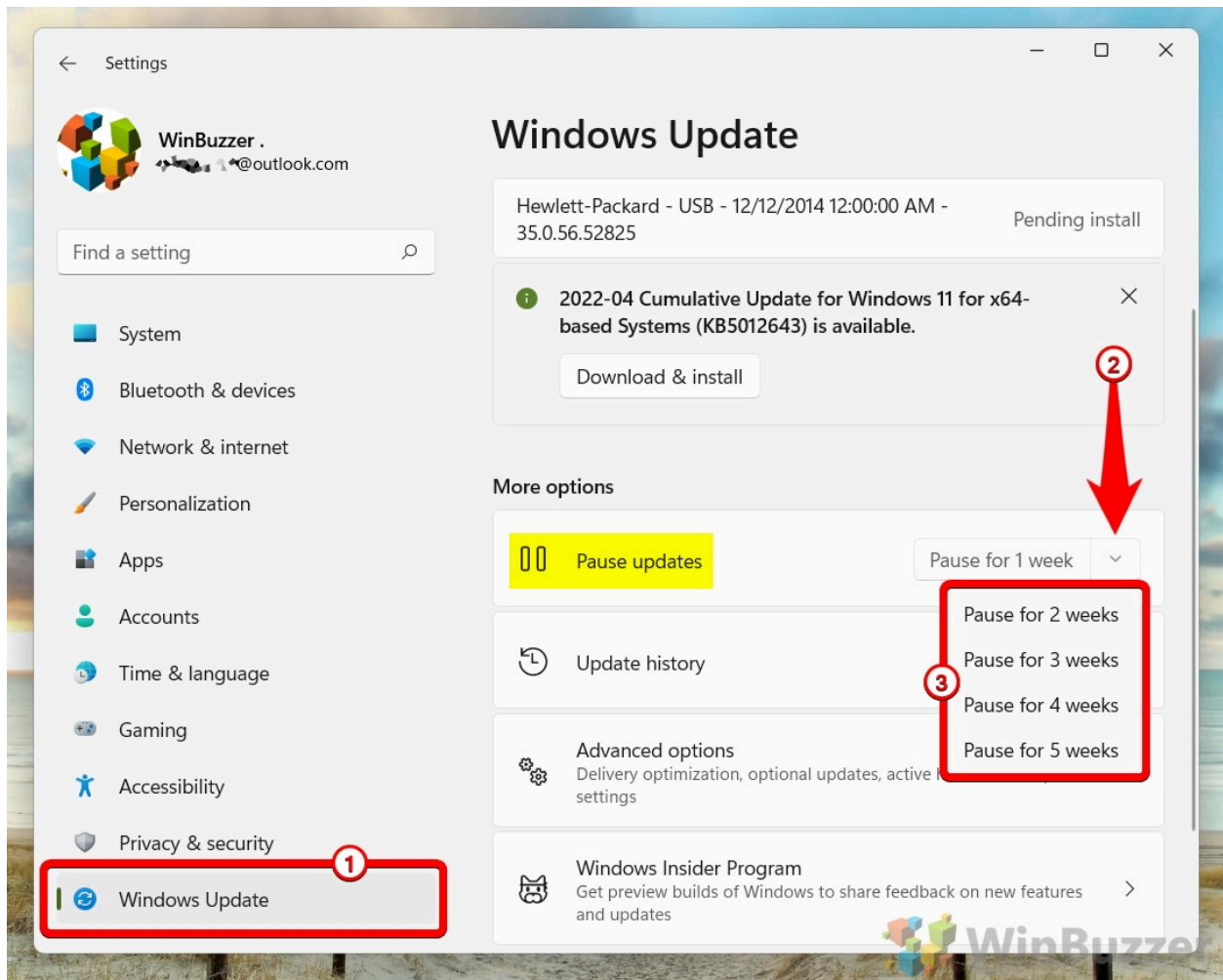
Keep Software Updated (Windows)

Keeping Windows updated is **very important** for security, bug fixes, and new features.

How to set automatic updates (recommended):

1. Go to **Settings** (Win + I)
2. Click **Windows Update**
3. Make sure **Receive updates for other Microsoft products** is on
4. Click **Advanced options** → Turn on **Get the latest updates as soon as they're available**
5. Windows will automatically download & install updates (you can pause for a while if needed)

Here are visual examples of the Windows Update settings screen:



winbuzzer.com

How to Turn off Automatic Updates in Windows 11 or Windows 10 ...